

A Uniformly Convex Set with an Extreme Point That Is Not Strongly Exposed

FA/Banach run fa.banach_001

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Status

This packet gives a counterexample to the question in Remark 4.15 of Garcia-Lirola–Grelier–Rueda Zoca [1]: in general, an extreme point of a uniformly convex set need not be strongly exposed by an ambient continuous functional.

Source Question

The source paper asks whether every extreme point is a strongly exposed point in a uniformly convex set K . The authors then prove that every such point is intrinsically strongly exposed by a functional continuous for the Minkowski functional $|\cdot|_K$, while noting that this functional need not be continuous for the ambient norm.

Definitions Used

We use the source paper’s definition: a symmetric bounded closed convex set K in a Banach space X is uniformly convex if for every $\varepsilon > 0$ there is $\delta > 0$ such that, whenever $x, y \in K$ and $\|x - y\| > \varepsilon$, one has

$$\frac{x + y}{2} \in (1 - \delta)K.$$

A strongly exposed point must in particular be exposed by some functional in the ambient dual X^* .

Proof Intuition

The source remark identifies the possible obstruction: the Minkowski functional of K may define a stronger Banach norm on $\text{span } K$ than the ambient norm inherited from X . The counterexample makes this gap explicit. Take a Hilbert ball for a stronger weighted ℓ_2 -norm and view it inside ordinary ℓ_2 . Its boundary points are extreme and the set is uniformly convex, but some Hilbert norming functionals are not continuous for the ambient ℓ_2 -norm. Such a point is intrinsically strongly exposed, exactly as in the source remark, but not exposed in the original Banach space.

Theorem 1. *There is a uniformly convex set K in a Banach space X and a point $x_0 \in \text{ext}(K)$ such that x_0 is not strongly exposed.*

Remark 4.15. In general we do not know whether every extreme point is a strongly exposed point in a uniformly convex set K . However, it turns out that every extreme point is strongly exposed in a sense which depends on the Minkowski functional $|\cdot|_K$. We say that a point is *intrinsically strongly exposed* if there exists a linear functional $f: X \rightarrow \mathbb{R}$ (not necessarily bounded) so that $f(x) = \sup_K f = 1$ and that, for every $\varepsilon > 0$, there exists $\delta > 0$ so that $f(y) > 1 - \delta \Rightarrow \|x - y\| < \varepsilon$.

Let us prove that if $x \in K$ satisfies that $|x|_K = 1$ then it is intrinsically strongly exposed. Define $Y = (\text{span}(K), |\cdot|_K)$. It is well known that Y is a Banach space such that $B_Y = K$. By the Hahn-Banach theorem, there exists $f \in S_{Y^*}$ such that $f(x) = |x|_K = 1$. Let us prove that f strongly exposes x in the above sense. To do so, pick $\varepsilon > 0$ and take δ of the definition of uniformly convex set. Now if $y \in K$ satisfies that $f(y) > 1 - \delta$ we get

$$1 - \frac{\delta}{2} < \frac{f(x+y)}{2} \leq \frac{|x+y|_K}{2} = \left| \frac{x+y}{2} \right|_K,$$

which means that $\frac{x+y}{2} \notin (1 - \delta)K$. This implies that $\|x - y\| < \varepsilon$. The arbitrariness of ε implies that every point of $\{x \in K : |x|_K = 1\}$ is strongly exposed (actually, it is uniformly strongly exposed).

Note that f is $|\cdot|_K$ -continuous. However, in the general case, f is not necessarily $\|\cdot\|$ -continuous (unless $\text{span}(K)$ is closed since in that case the norm induced by X on Y and $|\cdot|_K$ are equivalent by the open mapping theorem).

Remark 4.16. In spite of the previous remark, we can at least prove that $\text{str-exp}(K)$ is dense in $\text{ext}(K)$ in a uniformly convex set K . Indeed, since K is weakly compact, $K = \overline{\text{co}}(\text{str-exp}(K))$ and so $\text{dent}(K) \subset \overline{\text{str-exp}(K)}$. By Proposition 4.12, $\text{ext}(K) = \text{dent}(K)$ provided that K is uniformly convex.

Figure 1: Source evidence from arXiv:2109.01393, PDF page 26, Remark 4.15.

Proof. Work over the real scalar field and let $X = \ell_2$ with its usual norm. Define

$$K = \left\{ x = (x_n)_{n \geq 1} \in \ell_2 : \sum_{n=1}^{\infty} n^2 x_n^2 \leq 1 \right\}.$$

This set is symmetric, bounded, and convex. It is closed in ℓ_2 : if $x^{(m)} \rightarrow x$ in ℓ_2 and $\sum n^2 (x_n^{(m)})^2 \leq 1$ for all m , then for every N ,

$$\sum_{n=1}^N n^2 x_n^2 = \lim_{m \rightarrow \infty} \sum_{n=1}^N n^2 (x_n^{(m)})^2 \leq 1,$$

and letting $N \rightarrow \infty$ gives $x \in K$.

Let

$$|x|_K = \left(\sum_{n=1}^{\infty} n^2 x_n^2 \right)^{1/2}$$

on $\text{span } K$. This is a Hilbert norm and K is its unit ball. Since $\|x\|_2 \leq |x|_K$, if $u, v \in K$ and $\|u - v\|_2 > \varepsilon$, then $|u - v|_K > \varepsilon$. The unit ball of a Hilbert space is uniformly convex, so there is

$\delta > 0$ such that

$$\left| \frac{u+v}{2} \right|_K \leq 1 - \delta.$$

Equivalently, $(u+v)/2 \in (1-\delta)K$. Hence K is uniformly convex as a subset of the ambient Banach space $X = \ell_2$.

Choose

$$c = \left(\sum_{n=1}^{\infty} \frac{1}{n^2} \right)^{-1/2}, \quad x_0 = \left(\frac{c}{n^2} \right)_{n \geq 1}.$$

Then

$$|x_0|_K^2 = \sum_{n=1}^{\infty} n^2 \frac{c^2}{n^4} = c^2 \sum_{n=1}^{\infty} \frac{1}{n^2} = 1.$$

Because K is the unit ball of a strictly convex Hilbert norm, every $|\cdot|_K$ -unit vector is an extreme point of K . Thus $x_0 \in \text{ext}(K)$.

It remains to prove that x_0 is not exposed by any functional in $X^* = \ell_2$. Let $b = (b_n) \in \ell_2$, and let

$$\varphi_b(x) = \sum_{n=1}^{\infty} b_n x_n$$

be the corresponding ambient continuous functional. On the Hilbert space $(\text{span } K, |\cdot|_K)$, this functional is represented by the vector

$$z = \left(\frac{b_n}{n^2} \right)_{n \geq 1}, \quad \varphi_b(y) = \sum_{n=1}^{\infty} n^2 z_n y_n = \langle y, z \rangle_K.$$

If φ_b exposed x_0 over K , then in particular it would attain its maximum over the Hilbert unit ball K at x_0 . Equality in the Cauchy–Schwarz inequality for the Hilbert norm forces

$$z = \lambda x_0$$

for some $\lambda > 0$. Therefore

$$b_n = n^2 z_n = \lambda n^2 x_{0,n} = \lambda c$$

for every n . This coefficient sequence is constant and nonzero, hence it does not belong to ℓ_2 , a contradiction.

So x_0 is not exposed by any ambient continuous functional. Since every strongly exposed point is exposed, x_0 is not strongly exposed. This finishes the counterexample. $\square$

Verification Notes

The proof uses no computational verifier. The checks are structural:

- K is closed by finite-coordinate truncation and monotone convergence.
- Uniform convexity follows from the Hilbert unit ball for the stronger norm $|\cdot|_K$ and the inequality $\|x\|_2 \leq |x|_K$.
- The chosen point $x_0 = (c/n^2)$ lies on the $|\cdot|_K$ -unit sphere, hence is extreme.
- Ambient exposing functionals correspond to coefficient sequences in ℓ_2 . The only possible Hilbert norming coefficient sequence for x_0 is constant, hence not in ℓ_2 .

Novelty and Search Bounds

Before packaging, the run’s cheap indexes were searched for the arXiv id and the core terms “uniformly convex set”, “strongly exposed point”, and “extreme point”. The local parsed arXiv corpus was searched for close variants including “uniformly convex set strongly exposed”, “extreme point not strongly exposed”, “compact ellipsoid”, and “Minkowski functional strongly exposed”. A web search on June 14, 2026 for the exact question and close variants surfaced the source paper itself as the only relevant hit. No prior run packet or direct literature answer for this counterexample was found.

Human Review Recommendation

Review as a short counterexample packet. The key point for verification is the no-exposure argument: if an ambient ℓ_2 functional exposed x_0 , its weighted-Hilbert representative would have to be a positive scalar multiple of x_0 , forcing the ambient coefficient sequence to be constant and nonzero.

References

- [1] L. C. Garcia-Lirola, G. Grelier, and A. Rueda Zoca, *Extremal structure in ultrapowers of Banach spaces*, arXiv:2109.01393.

Human Review: Uniformly convex ellipsoid with non-exposed extreme point

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears correct and clean. It gives a counterexample to the question in Remark 4.15 of Garcia-Lirola–Grelier–Rueda Zoca, and in fact proves a slightly stronger statement: the chosen point is not merely not strongly exposed by an ambient continuous functional; it is not exposed at all by such a functional.

The construction makes explicit the phenomenon already suggested in the source remark: the Minkowski functional of K may define a stronger norm than the ambient norm. The point is intrinsically strongly exposed for the Minkowski norm, but the corresponding norming functional is not continuous for the ambient Banach-space norm.

The Example

The ambient space is $X = \ell_2$, and

$$K = \left\{ x = (x_n) \in \ell_2 : \sum_{n=1}^{\infty} n^2 x_n^2 \leq 1 \right\}.$$

The Minkowski functional is the weighted Hilbert norm

$$|x|_K = \left(\sum_{n=1}^{\infty} n^2 x_n^2 \right)^{1/2}.$$

Thus K is the unit ball of a Hilbert norm on $\text{span } K$. Since

$$\|x\|_2 \leq |x|_K,$$

uniform convexity of the Hilbert ball implies uniform convexity of K as a subset of the ambient space ℓ_2 .

The chosen point is

$$x_0 = \left(\frac{c}{n^2} \right)_{n \geq 1}, \quad c = \left(\sum_{n=1}^{\infty} \frac{1}{n^2} \right)^{-1/2}.$$

Then $|x_0|_K = 1$, so x_0 is an extreme point of K , because K is the unit ball of a strictly convex Hilbert norm.

Why It Is Not Exposed

Let an ambient continuous functional be represented by $b = (b_n) \in \ell_2$:

$$\varphi_b(x) = \sum_{n=1}^{\infty} b_n x_n.$$

With respect to the weighted Hilbert inner product

$$\langle x, z \rangle_K = \sum_{n=1}^{\infty} n^2 x_n z_n,$$

this functional is represented by

$$z = \left(\frac{b_n}{n^2} \right)_{n \geq 1}.$$

If φ_b exposed x_0 over K , then equality in the Hilbert-space Cauchy-Schwarz inequality would force

$$z = \lambda x_0$$

for some $\lambda > 0$. Since $x_{0,n} = c/n^2$, this implies

$$\frac{b_n}{n^2} = \lambda \frac{c}{n^2}, \quad \text{hence} \quad b_n = \lambda c$$

for every n . But a nonzero constant sequence is not in ℓ_2 . Thus no ambient continuous functional can expose x_0 .

Therefore x_0 is not exposed, and hence is not strongly exposed.

Review Outcome

Approve this packet as an apparently correct counterexample. The proof is short, transparent, and directly illustrates the distinction between exposure for the intrinsic Minkowski functional and exposure by a functional continuous for the ambient norm.